

SOFTWARE REQUIREMENT SPECIFICATION- HOSPITAL MANAGEMENT SYSTEM

1.Introduction

1.1 Purpose:

The purpose of the Hospital Management System is to provide a centralized platform for managing hospital operations, including patient registration, appointments, medical records, billing, and staff information. The system aims to improve operational efficiency, reduce manual errors, and ensure secure and accurate management of healthcare data.

1.2 Scope:

The Hospital Management System is designed to automate and simplify the day-to-day operations of a hospital. It provides features for patient registration, appointment scheduling, doctor and staff management, medical record maintenance, billing, and report generation. The system enables authorized users to securely access and manage hospital information, improving efficiency, accuracy, and the overall quality of healthcare services.

1.3 Objectives:

- To automate and simplify the daily operations of the hospital.
- To maintain accurate and secure records of patients, doctors, appointments, and billing information.
- To reduce paperwork, manual errors, and processing time through digital record management.
- To provide quick and easy access to patient and hospital information for authorized users.
- To improve the overall efficiency, reliability, and quality of hospital management services.

2. Overall Description

2.1 Product Perspective:

The Hospital Management System is a standalone application that automates hospital operations such as patient management, appointments, billing, and medical records. It provides a centralized and secure platform to improve the efficiency and accuracy of hospital services.

2.2 Product Functions:

- Register and manage patient information.
- Schedule and manage doctor appointments.
- Maintain doctors' and staff records.
- Store and update patients' medical records.

- Generate bills and manage payments.
- Produce reports for hospital administration.

2.3 User Classes:

- Administrator – Manages users, hospital records, and system settings.
- Doctor – Views patient records, updates diagnoses, and manages appointments.
- Receptionist – Registers patients, schedules appointments, and handles billing.
- Patient – Views appointment details, medical records, and billing information (if applicable).

2.4 Abbreviations:

- HMS – Hospital Management System
- SRS – Software Requirements Specification
- UI – User Interface
- DB – Database
- ID – Identification
- OTP – One-Time Password
- EMR – Electronic Medical Record

3. Functional Requirements

FR1: User Login

- The system shall allow authorized users to log in securely using a username and password.

FR2: Patient Management

- The system shall allow users to add, update, delete, and view patient records.

FR3: Doctor Management

- The system shall allow the administrator to manage doctor information and schedules.

FR4: Appointment Management

- The system shall allow patients to book appointments and staff to schedule, update, or cancel appointments.

FR5: Medical Records

- The system shall store and maintain patients' medical history, diagnoses, prescriptions, and treatment details.

FR6: Billing and Payment

- The system shall generate bills, record payments, and maintain billing history.

FR7: Staff Management

- The system shall maintain records of hospital staff and their roles.

FR8: Report Generation

- The system shall generate reports related to patients, appointments, billing, and hospital activities.

FR9: Search Function

- The system shall allow users to search for patients, doctors, and appointments using relevant details.

FR10: Logout

- The system shall allow users to securely log out of the application.

4. Non-Functional Requirements

NFR1: Performance

- The system shall provide quick responses and handle multiple user requests efficiently.

NFR2: Security

- The system shall ensure secure access through authentication and protect patient data from unauthorized access.

NFR3: Reliability

- The system shall provide accurate results and maintain consistent availability.

NFR4: Usability

- The system shall have a simple and user-friendly interface for easy operation.

NFR5: Maintainability

- The system shall be easy to update, modify, and maintain.

NFR6: Scalability

- The system shall support future expansion with additional features and increased data volume.

NFR7: Data Backup

- The system shall provide backup and recovery mechanisms to prevent data loss.

5. Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Storage: Minimum 500 GB Hard Disk / SSD
- Monitor: Standard display with minimum 1024 × 768 resolution

- Input Devices: Keyboard and mouse
- Network: LAN/Wi-Fi connection for data access and communication

6. Software Requirements

- Operating System: Windows 10/11 or Linux
- Frontend: HTML, CSS, JavaScript (or any suitable UI framework)
- Backend: Java / Python / PHP (based on implementation)
- Database: MySQL or any relational database management system
- Development Environment: Visual Studio Code / Eclipse / IntelliJ IDEA
- Web Browser: Google Chrome or any modern browser

7. Database Requirements

- The system shall use a relational database to store and manage hospital-related information.
- The database shall maintain records of patients, doctors, appointments, medical history, billing, and staff details.
- The database shall ensure data accuracy, integrity, and security through proper access control.
- The system shall support data retrieval, updating, and backup operations efficiently.

8. Assumptions and Constraints

Assumptions

- Users have basic knowledge of computers and can operate the system.
- Authorized users have valid login credentials to access the system.
- The hospital has the required hardware and network facilities to run the application.
- Patient and hospital data provided to the system is accurate.

Constraints

- The system requires proper authentication to access sensitive information.
- The application depends on database availability for storing and retrieving data.
- Only authorized users can modify or delete hospital records.
- System performance may depend on available hardware and network resources.

9. Future Enhancements

- Integration of online appointment booking and video consultation features.
- Implementation of mobile application support for patients and doctors.
- Addition of AI-based diagnosis assistance and health recommendations.
- Integration with online payment gateways for secure transactions.

- Implementation of automated notifications for appointments, reports, and reminders.
- Enhancement of data analytics for generating healthcare insights and reports.

10. Conclusion

The Hospital Management System provides an efficient and reliable solution for managing hospital operations digitally. It simplifies patient management, appointments, medical records, and billing processes while improving accuracy, security, and overall healthcare service efficiency.